Predicting Squirrel–Human Interaction Behaviour

A data science and machine learning project that investigates whether squirrel behaviour toward humans can be predicted using behavioural, spatial, temporal, and environmental features derived from the 2018 Central Park Squirrel Census dataset.

Overview

Urban wildlife regularly interacts with humans, yet the factors influencing these interactions are often complex and interconnected. This project explores whether squirrels in Central Park, New York City, are likely to:

- Approach humans
- Avoid humans
- Remain indifferent

Using data preprocessing, feature engineering, correlation analysis, supervised machine learning, and clustering techniques, the project examines how behaviour, environment, location, and time influence squirrel–human interactions.

Research Question

> Can we predict whether a squirrel will approach or avoid humans based on its behaviour, location, and time of observation, and which features are most influential?

Dataset

The project uses data from the **2018 Central Park Squirrel Census** along with additional environmental information.

Data Sources

File	Description
`squirrel.csv`	Main squirrel observation dataset
`hectare.csv`	Environmental and area-level information
`stories.csv`	Supplementary squirrel observation records
`task_1_preprocessed_squirrel_data.csv`	Cleaned and engineered dataset used for modelling

Environmental Data Integrated

- Crowd conditions

- Presence of threat animals
- Total squirrels observed per area
- Geographic location within Central Park

Project Pipeline

```
Raw Datasets
      │
      ▼
Data Preprocessing
      │
      ▼
Feature Engineering
      │
      ▼
Correlation Analysis
      │
      ▼
Supervised Learning
(Logistic Regression & Random Forest)
      │
      ▼
K-Means Clustering
      │
      ▼
Behavioural Insights
```

Data Preprocessing

Feature Engineering

Several behavioural variables were aggregated into higher-level behavioural scores.

Behaviour Scores

Feature	Description
`active_score`	Running + Chasing + Climbing
`foraging_score`	Eating + Foraging

Feature	Description
`vocal_score`	Kuks + Quaas + Moans
`alert_score`	Tail Flags + Tail Twitches

Temporal Features

Feature	Description
`shift_enc`	AM = 0, PM = 1
`weekday`	Day of week encoded as 0–6

Environmental Features

Feature	Description
`conditions_enc`	Encoded crowd conditions
`has_threat`	Presence of dogs or cats
`total_squirrels`	Number of squirrels observed in area

Spatial Features

Feature	Description
`X`	Longitude
`Y`	Latitude
`zone`	NE, NW, SE, SW park regions
`density_level`	Low, Medium, High squirrel density

Missing Value Handling

Variable	Strategy
Age	Mode Imputation
Primary Fur Color	Mode Imputation
Location	Mode Imputation
total_squirrels	Fill with 0
has_threat	Fill with 0
conditions_enc	Mode Imputation

Variable	Strategy
Highlight Fur Color	Removed

Target Variable

The original interaction columns were combined into a multiclass target:

Class	Behaviour
0	Runs Away
1	Indifferent
2	Approaches

This transformation prevents target leakage during model training.

Correlation Analysis

Pearson Correlation was used to evaluate relationships between squirrel behaviour and human interaction outcomes.

Features Analysed

Behavioural Features

- Running
- Chasing
- Climbing
- Eating
- Foraging
- Kuks
- Quaas
- Moans
- Tail Flags
- Tail Twitches
- Engineered behaviour scores

Environmental Features

- Total squirrels
- Threat presence
- Crowd conditions

Spatial Features

- X coordinate
- Y coordinate

Temporal Features

- Observation shift

Key Findings

Positive Correlations

- `foraging_score`
- `Foraging`
- `Eating`

Food-seeking squirrels were more likely to interact positively with humans.

Negative Correlations

- `Running`
- `active_score`
- `X`
- `Y`

More active squirrels tended to avoid human interaction.

Overall Conclusion

No feature showed a strong standalone correlation with squirrel–human interaction behaviour, suggesting behaviour is influenced by multiple interacting factors rather than a single dominant predictor.

Supervised Learning

Two classification models were developed to predict squirrel reactions toward humans.

Features Used

Behavioural

- active_score
- foraging_score
- vocal_score
- alert_score

Temporal

- shift_enc
- weekday

Environmental

- zone
- density_level
- conditions_enc
- has_threat
- outlier_variable

Spatial

- X
- Y

Squirrel Characteristics

- Age
- Primary Fur Color
- Location

Logistic Regression

Configuration

```
LogisticRegression(
    max_iter=2000,
    class_weight="balanced"
)
```

Pipeline

- Mean imputation (numeric)
- Most-frequent imputation (categorical)
- StandardScaler
- OneHotEncoder
- 5-fold cross validation

Performance

Metric	Score
Accuracy	42.8%

Metric	Score
Precision	65.8%
Recall	42.8%
Weighted F1	48.0%

Strengths

- Better minority-class detection
- More interpretable coefficients
- More balanced predictions

Random Forest

Configuration

```
RandomForestClassifier(
    n_estimators=200,
    max_depth=12,
    min_samples_split=5,
    random_state=42
)
```

Performance

Metric	Score
Accuracy	73.6%
Precision	66.3%
Recall	73.6%
Weighted F1	66.7%

Strengths

- Captured nonlinear relationships
- Higher predictive performance
- Better overall classification accuracy

Limitations

- Strong bias toward the majority class
- Failed to classify the "Approaches" category effectively

Most Important Features (Random Forest)

Top predictors identified:

1. Y Coordinate
2. X Coordinate
3. Weekday
4. active_score
5. foraging_score
6. conditions_enc
7. alert_score
8. shift_enc
9. has_threat
10. density_level

This suggests location plays a major role in squirrel–human interactions.

Clustering Analysis

To discover hidden behavioural and environmental patterns, two separate K-Means clustering analyses were performed.

Behaviour Clustering

Features

- active_score
- foraging_score
- vocal_score
- alert_score

Scaling

```
MinMaxScaler()
```

Cluster Selection

Elbow Method:

```
Optimal K = 5
```

Behaviour Clusters

Cluster	Interpretation
Cluster 0	Forager
Cluster 1	Active
Cluster 2	Alert Forager
Cluster 3	Most Forager
Cluster 4	Alert Active

Findings

- Alert squirrels showed higher avoidance rates.
- Strong foragers displayed the highest approach rates.
- Indifferent behaviour dominated all clusters.

Environment Clustering

Features

- X
- Y
- conditions_enc
- has_threat
- total_squirrels

Scaling

```
StandardScaler()
```

Cluster Selection

```
Optimal K = 3
```

Environment Clusters

Cluster	Interpretation
Cluster 0	High-Threat Low-Density
Cluster 1	Medium Environment

Cluster	Interpretation
Cluster 2	High-Density Busy

Findings

- Medium Environment clusters showed the highest avoidance behaviour.
- High-Density Busy areas contained the largest squirrel populations.
- Human interaction patterns varied geographically across the park.

Results

Major Findings

1. Location Matters Most

Spatial coordinates (X and Y) consistently emerged as the strongest predictors of squirrel–human interactions.

2. Foraging Behaviour Encourages Interaction

Squirrels engaged in food-seeking behaviour were more likely to approach or tolerate humans.

3. Alert Behaviour Increases Avoidance

Alert and highly active squirrels were more likely to avoid humans.

4. Neutral Behaviour Dominates

Approximately 72% of observations involved squirrels showing indifferent behaviour toward humans.

5. Behaviour Is Multifactorial

No single feature explains squirrel–human interactions. Instead, behaviour appears to be influenced by behavioural, environmental, spatial, and temporal factors together.

Repository Structure

```
.
├── squirrel.csv
├── hectare.csv
├── stories.csv
```

```
│
├── task_1_data_preprocessing.py
├── task_1_preprocessed_squirrel_data.csv
│
├── task_2_correlation.py
├── task_2_distribution_of_human_interaction_categories.jpg
├── task_2_feature_correlations_with_human_interaction_behaviour.jpg
├── task_2_target_correlations.csv
│
├── task_3_supervised_machine_learning_models.py
├── task_3_random_forest_feature_importance.csv
├── task_3_logistic_regression_coefficients.csv
│
├── task_4_clustering.py
├── task4_elbow_1.png
├── task4_elbow_2.png
├── task4_heatmap_feat_mean_1.png
├── task4_heatmap_feat_mean_2.png
├── task4_human_reaction_1.png
├── task4_human_reaction_2.png
├── task4_scatter_env_clusters.png
│
└── report.pdf
```

Installation

```
git clone <repository-url>
cd squirrel-human-interaction
```

Install required dependencies:

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

Usage

Run preprocessing:

```
python task_1_data_preprocessing.py
```

Run correlation analysis:

```
python task_2_correlation.py
```

Run supervised learning models:

```
python task_3_supervised_machine_learning_models.py
```

Run clustering analysis:

```
python task_4_clustering.py
```

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Machine Learning

- Logistic Regression
- Random Forest
- K-Means Clustering

Data Processing

- Feature Engineering
- Missing Value Imputation
- Outlier Detection
- One-Hot Encoding
- Feature Scaling
- Correlation Analysis

Limitations

- Significant class imbalance, particularly for the "Approaches" class.
- Pearson correlation only captures linear relationships.
- K-Means assumes simple cluster structures.
- Limited minority-class observations reduced predictive performance.
- Random Forest struggled with minority-class classification.

Future Improvements

- Collect larger and more balanced datasets.
- Apply oversampling methods such as SMOTE.
- Explore gradient boosting models (XGBoost, LightGBM).
- Incorporate weather and seasonal information.
- Perform advanced geospatial analysis.
- Investigate deep learning approaches for behavioural prediction.

Contributors

Group W14G02

- Alisina Nekounam
- Jinmo Joo
- Thesara Hirimuthugoda
- Shizhen Jia

References

- 2018 Central Park Squirrel Census Dataset
- NYC Open Data
- Python Documentation
- Scikit-learn Documentation
- Pandas Documentation
- Seaborn Documentation

Conclusion

This project demonstrates that squirrel–human interaction behaviour can be partially predicted using behavioural, environmental, spatial, and temporal features. The results suggest that location, behavioural tendencies, and environmental context collectively influence how squirrels respond to humans in urban environments. Random Forest achieved the strongest predictive performance, while clustering analysis revealed meaningful behavioural and environmental groups that help explain interaction patterns.